

When changing the chart helps

Large numerical gains over mean bars

True difference Recovered from mean bars Recovered from selected chart

a Breast tumour / Fractal dimension (worst)

A: Benign (n=357)

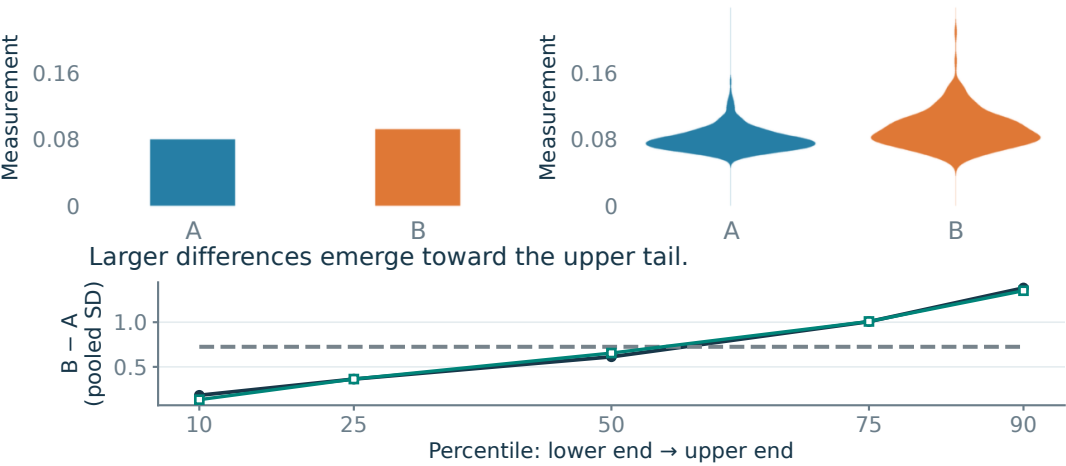
B: Malignant (n=212)

Mean bars 71.87

→ Violin plot 97.57

+25.70

score points



RANK / CHART		SCORE / 100
1	Violin plot 256 grid points	97.566
2	Dot plot raw observations	96.724
3	Heatmap 32 bins	96.540
4	Histogram 32 bins	96.278
5	Bar chart means	71.865
6	Interval plot five-number	61.220
7	Box plot five-number	61.126

b Smartphone activity / Gyroscope-jerk variability

A: Sitting (n=30)

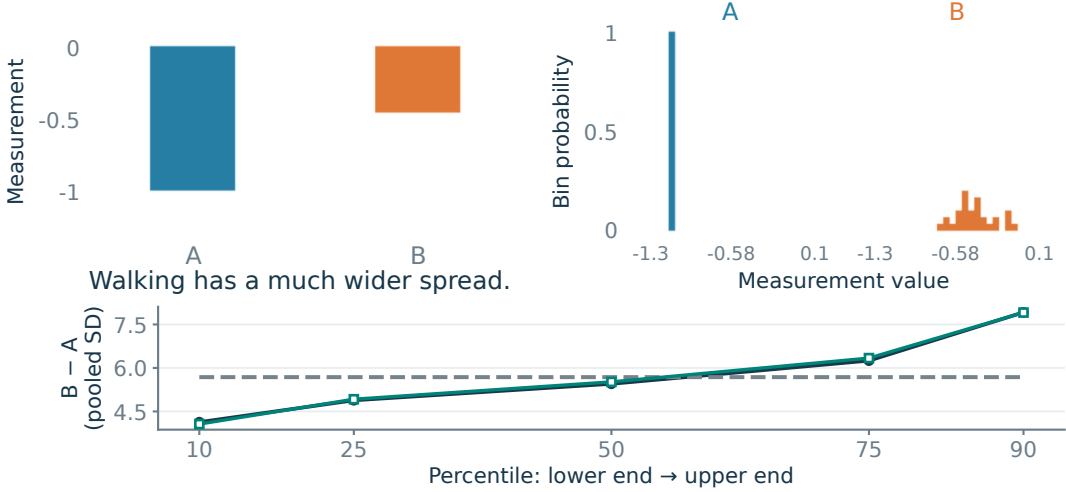
B: Walking (n=30)

Mean bars 48.15

→ Histogram 94.96

+46.81

score points



RANK / CHART		SCORE / 100
1	Histogram 32 bins	94.960
2	Heatmap 32 bins	93.974
3	Box plot five-number	92.907
4	Dot plot raw observations	90.020
5	Interval plot five-number	89.943
6	Violin plot 32 grid points	87.802
7	Bar chart means	48.148

c Heart disease / ST depression (oldpeak)

A: No disease (n=164)

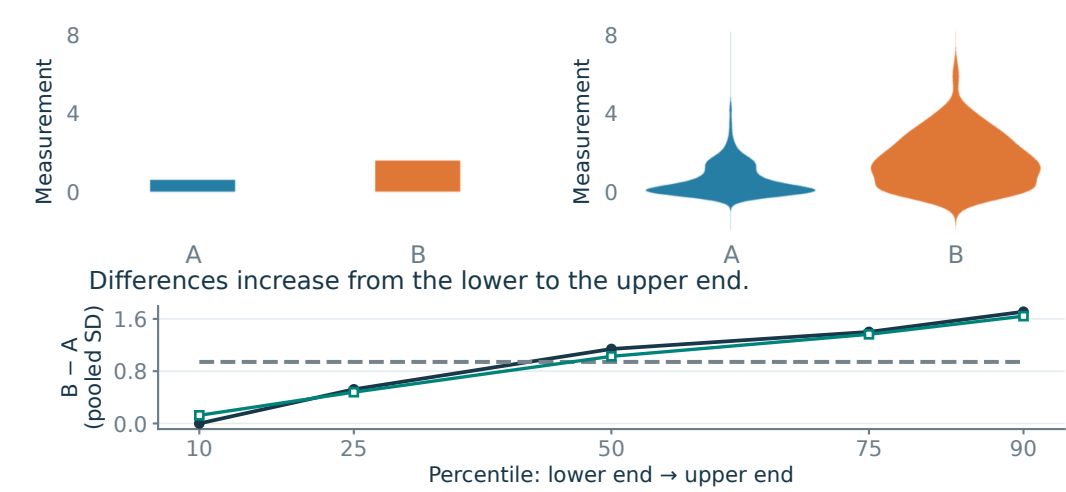
B: Disease (n=139)

Mean bars 64.22

→ Violin plot 92.84

+28.62

score points



RANK / CHART		SCORE / 100
1	Violin plot 256 grid points	92.837
2	Histogram 64 bins	92.809
3	Heatmap 64 bins	92.014
4	Dot plot raw observations	91.860
5	Interval plot five-number	76.182
6	Box plot five-number	75.923
7	Bar chart means	64.220